MCSC-106 • 3 CREDITS

Pattern Recognition

Learn representations and decision rules from experience, then evaluate what generalizes.

STUDY BLOCKS	FORMAT	SCOPE
9	Definitions Equations Worked examples	Supplied syllabus only

How to use this handbook

For every study block: define the terms, reproduce the relation and its assumptions, solve the example without looking, and explain when the method is unsuitable.

Scope note: This handbook expands only topics named in the supplied syllabus. Confirm examination guidance, prescribed texts, and institutional policy with the responsible instructor.

Course map

1. Learning problems and models

Introduction, Algorithmic models of learning, Classifiers, Functions, Relations, Grammars, Probabilistic models, Value functions, Behaviors, Programs from experience

2. Bayesian, MAP and MDL frameworks

Bayesian framework, Maximum a posteriori, Minimum description length

3. Parameter estimation and model families

Parameter estimation, Sufficient statistics, Decision trees, Neural networks, Support vector machines, Bayesian networks

4. Text and sequence models

Bag of words classifiers, N-gram models, Markov models, Hidden Markov models, Probabilistic relational models

5. Instance, rule and ensemble methods

Association rules, Nearest neighbour classifiers, Locally weighted regression, Ensemble classifiers

6. Computational learning theory

Mistake bound analysis, Sample complexity, VC dimension, Occam learning, Accuracy and confidence boosting

7. Representation, reduction and visualization

Dimensionality reduction, Feature selection, Visualization

8. Clustering and mixture models

Clustering, Mixture models, K-means, Hierarchical clustering, Distributional clustering

9. Document recognition application

Applications: document recognition

Study block 01

Learning problems and models

Introduction	Algorithmic models of learning
Classifiers	Functions
Relations	Grammars
Probabilistic models	Value functions
Behaviors	Programs from experience

Definition and idea

Learning uses experience to improve performance on a defined task. The output may be a label, function, relation, grammar, probability model, value estimate, behavior, or program. A sound formulation identifies input representation, target, loss, data source, and evaluation protocol.

```
empirical risk = (1/n) sum L(f(x_i), y_i)
```

Worked example

Document recognition maps an image or token sequence to a label. The same data could instead support probability estimation, structured parsing, or a learned decision policy.

MUST UNDERSTAND
Define task, experience, performance measure, and generalization target.

Study block 02

Bayesian, MAP and MDL frameworks

Bayesian framework	Maximum a posteriori
Minimum description length	

Definition and idea

Bayesian inference combines prior belief and likelihood into a posterior. MAP selects the parameter with highest posterior density. MDL favors the model that gives the shortest combined description of model and data.

```
posterior proportional to likelihood x prior; theta_MAP=argmax p(D|theta)p(theta)
```

Worked example

With limited data, a prior can stabilize parameter estimates. As data accumulate, the likelihood often dominates. MDL penalizes complexity when it does not compress the observations enough.

MUST UNDERSTAND
State priors and model assumptions; do not present MAP as prior-free.

Study block 03

Parameter estimation and model families

Parameter estimation	Sufficient statistics
Decision trees	Neural networks
Support vector machines	Bayesian networks

Definition and idea

Parameter estimation fits a chosen model family. A sufficient statistic preserves all sample information relevant to a parameter under the model. Trees partition feature space, neural networks compose learned transformations, SVMs maximize a margin, and Bayesian networks encode conditional dependence.

```
SVM margin is proportional to 1/||w||
```

Worked example

For a Bernoulli model, the number of successes is sufficient for the success probability. A decision tree split can be chosen by impurity reduction, then validated on unseen data.

MUST UNDERSTAND
Compare interpretability, capacity, optimization, data needs, and calibration.

Study block 04

Text and sequence models

Bag of words classifiers	N-gram models
Markov models	Hidden Markov models
Probabilistic relational models	

Definition and idea

Bag of words represents token counts while discarding order. N-grams preserve short local context. Markov models assume limited state dependence; hidden Markov models infer unobserved states from observed emissions. Relational models include dependencies among entities.

```
P(w_1...w_T) approximately product P(w_t | previous n-1 words)
```

Worked example

A bigram model estimates a word from its predecessor. In an HMM for document zones, hidden states might be title, paragraph, and caption while observations are visual or textual features.

MUST UNDERSTAND
Identify what information is lost by the independence or Markov assumption.

Study block 05

Instance, rule and ensemble methods

Association rules	Nearest neighbour classifiers
Locally weighted regression	Ensemble classifiers

Definition and idea

Association rules describe co-occurrence patterns using support and confidence. Nearest-neighbour methods predict from nearby training examples. Local regression fits around a query point. Ensembles combine diverse learners to reduce error.

```
confidence(A -> B)=support(A union B)/support(A)
```

Worked example

In k-nearest neighbours, feature scaling changes distance and therefore neighbours. A validation set can choose k; test data must remain untouched until final evaluation.

MUST UNDERSTAND
Check leakage, scaling, distance choice, diversity, and validation design.

Study block 06

Computational learning theory

Mistake bound analysis	Sample complexity
VC dimension	Occam learning
Accuracy and confidence boosting	

Definition and idea

Learning theory relates hypothesis complexity, data volume, errors, accuracy, and confidence. VC dimension measures the largest set a hypothesis class can shatter. Occam-style reasoning controls complexity; boosting combines weak learners into a stronger classifier.

```
generalization bounds commonly grow with complexity and shrink with sample size
```

Worked example

A more expressive classifier may fit training data better yet need more observations to guarantee similar test performance. Boosting repeatedly emphasizes misclassified cases.

MUST UNDERSTAND
Separate training error from a claim about unseen-data error.

Study block 07

Representation, reduction and visualization

Dimensionality reduction	Feature selection
Visualization	

Definition and idea

Feature selection retains a subset of original variables. Dimensionality reduction maps data to a smaller representation. Visualization supports exploration but a two-dimensional projection can distort global or local structure.

```
PCA chooses directions maximizing projected variance
```

Worked example

Standardize differently scaled features before PCA when scale should not dominate. Fit the transformation on training data only, then apply it to validation and test data.

MUST UNDERSTAND
Treat a visualization as evidence to investigate, not automatic proof of separable classes.

Study block 08

Clustering and mixture models

Clustering	Mixture models
K-means	Hierarchical clustering
Distributional clustering	

Definition and idea

Clustering groups observations without supplied labels. K-means minimizes within-cluster squared distance to centroids. Hierarchical methods build nested merges or splits. Mixture models view observations as arising from latent component distributions.

```
k-means objective = sum_i ||x_i - mu_(cluster(i))||^2
```

Worked example

Different initial centroids can give different k-means solutions, so use multiple starts and report stability. A Gaussian mixture gives soft membership probabilities rather than only hard labels.

> **MUST UNDERSTAND**
> Justify distance, scaling, cluster count, and evaluation method.

Study block 09

Document recognition application

Applications: document recognition	

Definition and idea

Document recognition integrates preprocessing, segmentation, representation, classification or sequence modeling, and evaluation. Errors can propagate across stages, so component and end-to-end metrics are both useful.

```
F1 = 2 precision recall / (precision + recall)
```

Worked example

A system may deskew a scan, detect text regions, recognize characters, reconstruct reading order, and compare output with ground truth using character or word error rate.

MUST UNDERSTAND
Document data provenance, train-test separation, class imbalance, and representative failure cases.

Revision checklist

Use this final pass before an assessment or interview:

- ■ I can define every listed syllabus term in plain language.
- ■ I can reproduce the central equations and state their assumptions.
- ■ I can complete each worked example and create one variation.
- ■ I can compare closely related methods and justify a choice.
- ■ I can name limitations, failure modes, and ethical or operational concerns where relevant.